

Estimating π with Monte Carlo sampling

PhD Artificial Intelligence | Matin Esmaeili | seed = 20260914 | runtime = 4.7 s

Code, figures and data: <https://github.com/MatinEsmaeili00/monte-carlo-pi-clt>

1. Method

Points (x, y) are drawn uniformly on the unit square. A point counts as a hit when $x^2 + y^2 \leq 1$, i.e. when it falls inside the quarter disc of radius 1. That quarter disc has area $\pi/4$ while the square has area 1, so the hit indicator I is Bernoulli with $p = \pi/4$ and the estimator is $\text{pihat}_n = 4 * (\text{hits} / n)$. Sampling is vectorised with NumPy and generated in blocks of one million points, so memory stays flat even for $n = 10^7$.

2. What the theory predicts

$E[\text{pihat}_n] = 4p = \pi$, so the estimator is unbiased at every n . Since $\text{Var}(4I) = 16 p (1 - p) = \pi (4 - \pi)$, the variance of the estimator is $\pi (4 - \pi) / n$ and its standard deviation is

$$\text{sd}(\text{pihat}_n) = \sqrt{\pi (4 - \pi) / n} = 1.6422 / \sqrt{n}$$

The Central Limit Theorem applies because pihat_n is an average of n i.i.d. bounded terms. Therefore, for large n ,

$$\text{pihat}_n \sim \text{Normal}(\pi, \pi (4 - \pi) / n)$$

Two consequences are tested below. First, the error shrinks like $1/\sqrt{n}$: reducing it by a factor of 10 costs 100 times more points, which is why Monte Carlo is a poor way to compute digits of π but a workable way to integrate in high dimensions, where the $1/\sqrt{n}$ rate is independent of dimension. Second, the histogram of repeated estimates at fixed n must look Gaussian, centred at π with the width given above.

3. Convergence with n

One independent estimate was computed for each of 25 sample sizes spaced logarithmically from 10 to 10,000,000 points.

n	estimate	abs. error	sd = 1.6422/sqrt(n)
10	3.20000	0.05841	0.51930
31	3.61290	0.47131	0.29494
100	2.92000	0.22159	0.16422
316	3.13924	0.00235	0.09238
1,000	3.11600	0.02559	0.05193
3,162	3.15497	0.01337	0.02920
17,782	3.15240	0.01081	0.01231
56,234	3.15133	0.00974	0.00693
177,827	3.13408	0.00751	0.00389
562,341	3.14128	0.00031	0.00219
1,778,279	3.14424	0.00264	0.00123
10,000,000	3.14141	0.00018	0.00052

24 of the 25 estimates fall inside the theoretical 95% envelope $\pi \pm 1.96 \text{ sd}(n)$ (about 95% expected). On the log-log plot the observed errors scatter around the $\sigma/\sqrt{n}$ line with the predicted slope of $-1/2$, confirming the square-root rate rather than any faster convergence.

4. Sampling distribution at fixed n

For each n the whole experiment was repeated $R = 500$ times with independent points, giving R estimates whose spread is the sampling distribution. Empirical moments are compared with the CLT prediction:

n	mean	bias	emp sd	CLT sd	ratio	skew	kurt	KS	cov
1,000	3.14126	-0.00034	0.05242	0.05193	1.009	-0.032	+0.397	0.028	0.948
10,000	3.14033	-0.00127	0.01749	0.01642	1.065	+0.005	-0.245	0.056	0.924
100,000	3.14176	+0.00017	0.00534	0.00519	1.028	+0.018	-0.063	0.035	0.946

KS is the Kolmogorov-Smirnov distance between the empirical distribution and the exact CLT limit $N(\pi, \sigma^2/n)$; at $R = 500$ the 5% critical value is 0.061, so a value below that is consistent with normality. 'kurt' is excess kurtosis and 'cov' the fraction of replicates whose 95% confidence interval contains π .

5. Reading the results

The empirical standard deviations track the predicted $\sqrt{\pi(4-\pi)/n}$ closely (observed / predicted = 1.009, 1.065, 1.028), and each mean sits within a couple of standard errors of π , confirming unbiasedness. Every tenfold increase in n shrinks the spread by $\sqrt{10} \sim 3.16$, which is exactly what the three histograms show: same shape, same centre, width divided by about 3.16 at each step.

Shape statistics agree with a normal law (skewness -0.032, +0.005, +0.018, excess kurtosis near zero), the Q-Q points lie on the 45-degree line apart from the usual tail wobble, and the KS distances stay under their critical value. Note that the underlying draws are Bernoulli - as discrete and non-normal as a variable can be - yet their average is already indistinguishable from Gaussian at $n = 1000$. That is the Central Limit Theorem doing the work: normality is a property of the average, not of the data.

6. Conclusion

The Monte Carlo estimator is unbiased, converges at the $1/\sqrt{n}$ rate predicted by its variance $\pi(4-\pi)/n$, and its sampling distribution is approximately $\text{Normal}(\pi, \pi(4-\pi)/n)$ for every n tested. The practical reading: n controls precision, and buying one more correct decimal digit costs a hundredfold more samples.

Hit-or-miss sampling, $n = 3,000$

$4 \times 2387/3000 = 3.1827$

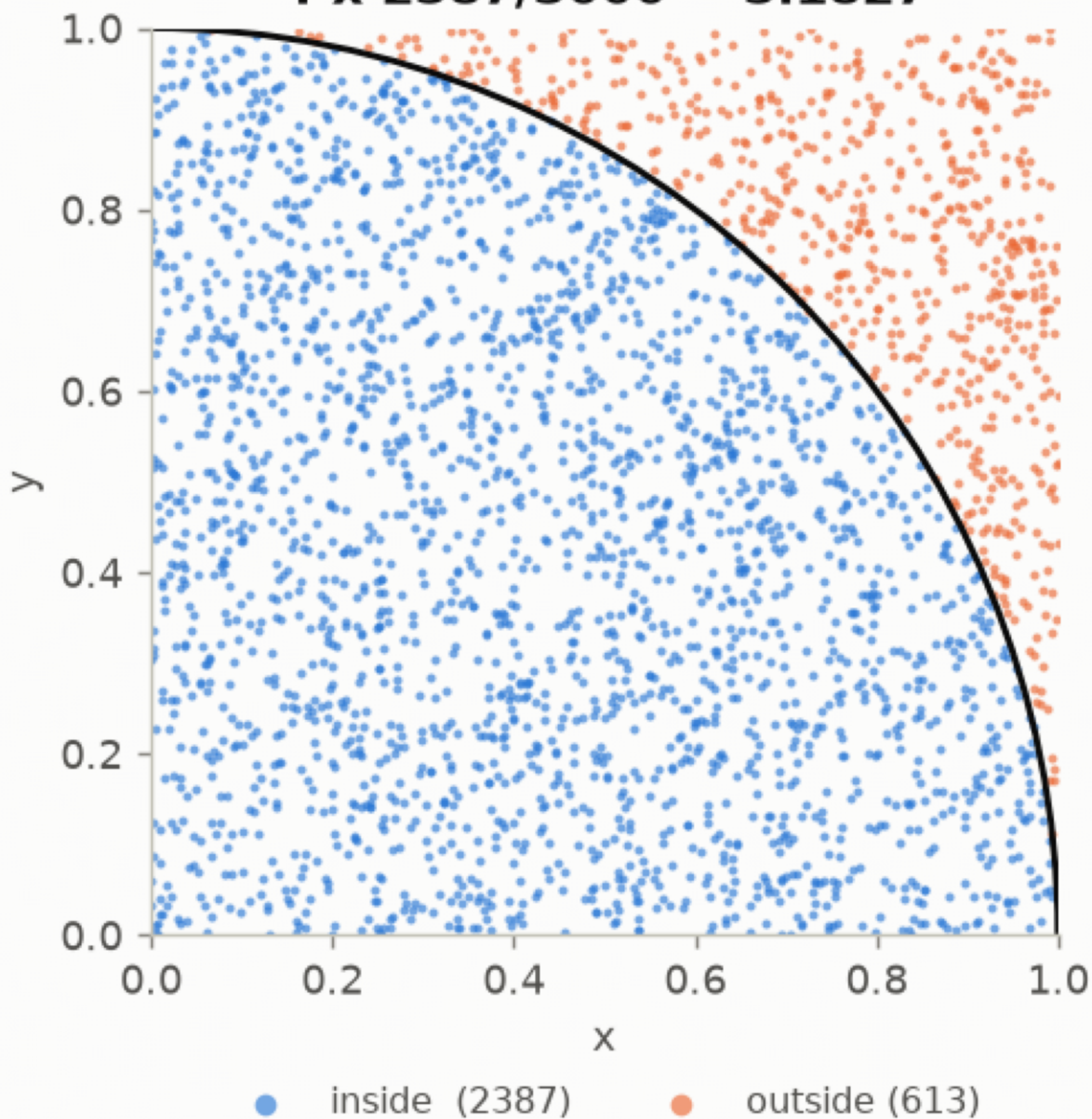


Figure 1. Hit-or-miss sampling. The fraction of points inside the quarter disc estimates $\pi/4$.

Convergence of the Monte Carlo estimate

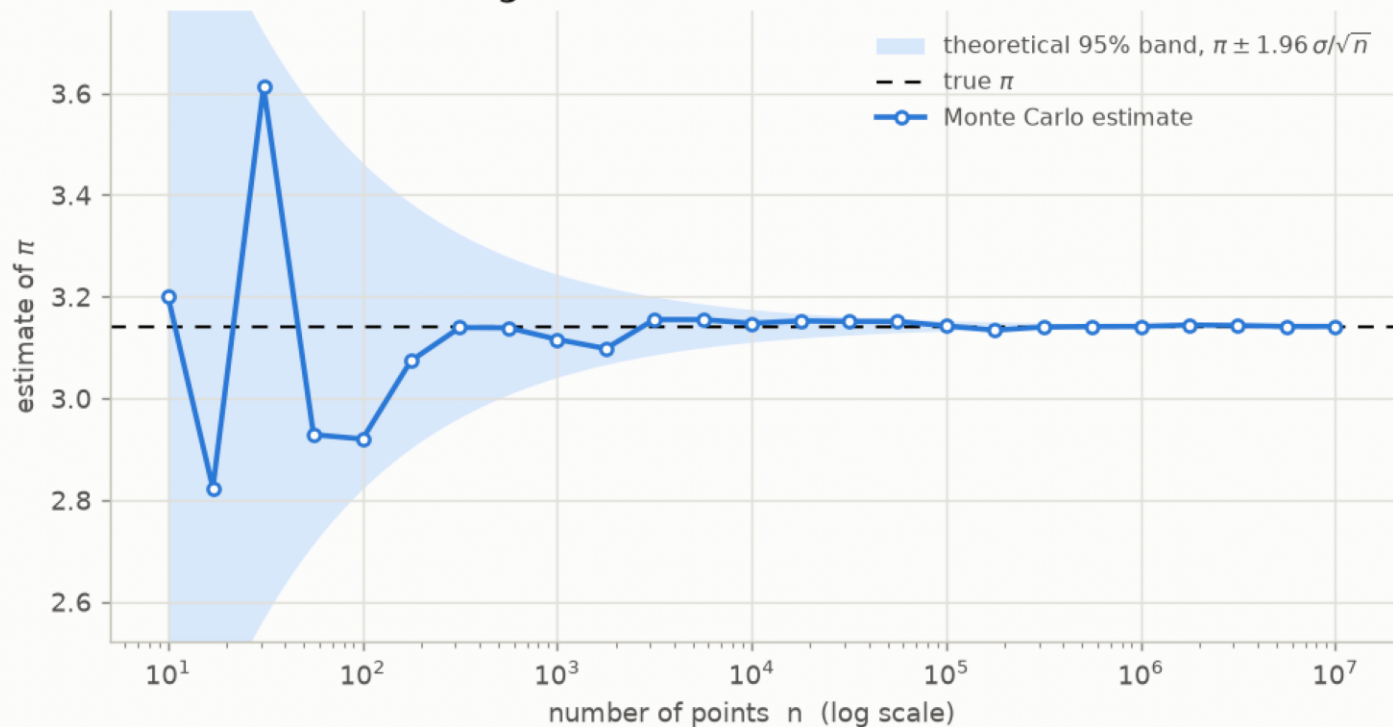


Figure 2. Estimate against n on a log x-axis. The band is the theoretical 95% envelope, not a fit to the data.

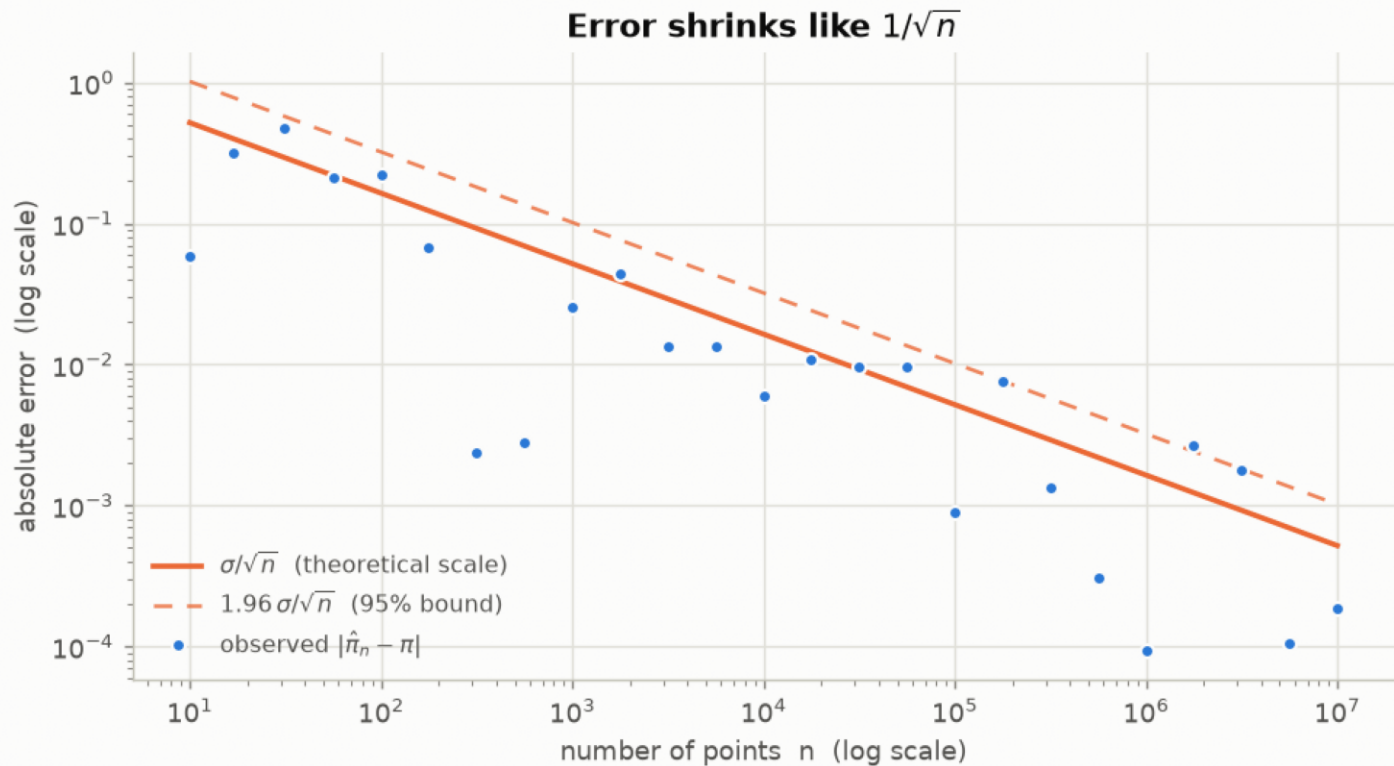


Figure 3. Absolute error against n on log-log axes; the slope matches the predicted $-1/2$.

Sampling distribution of $\hat{\pi}_n$ over independent repetitions

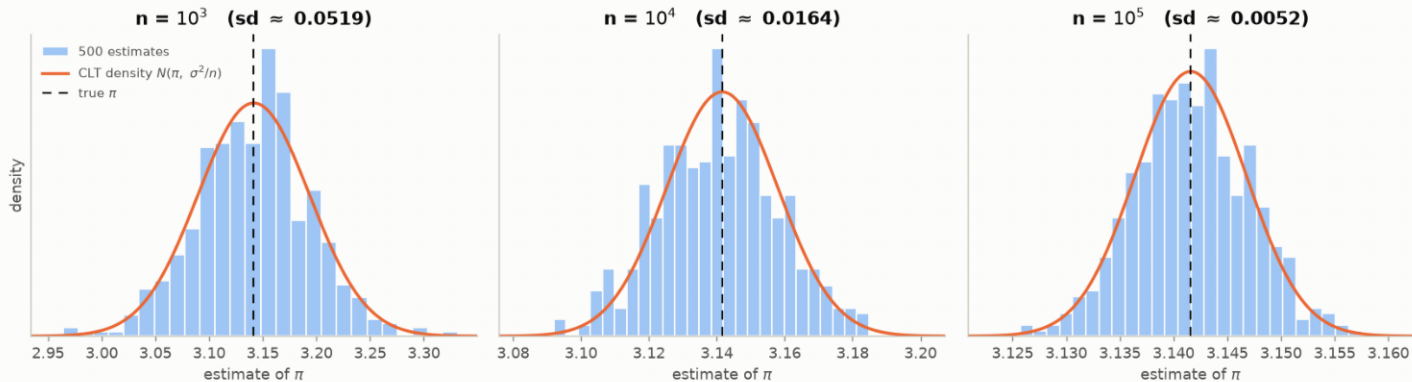


Figure 4. Sampling distribution at three values of n with the CLT density overlaid (not fitted).

Normal Q-Q plots of the sampling distribution

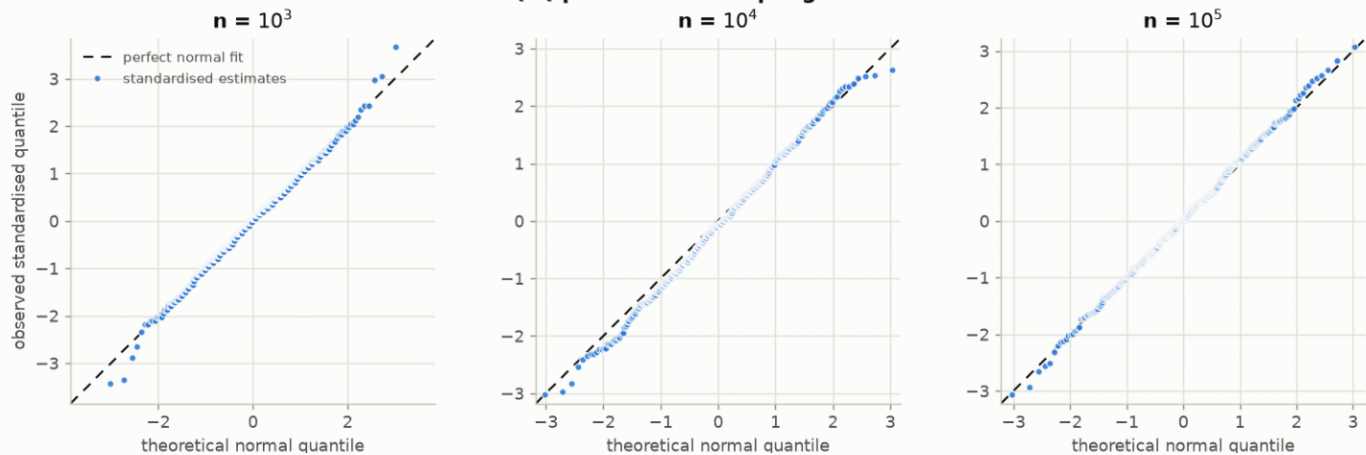


Figure 5. Normal Q-Q plots. Points on the diagonal indicate normality of the estimator.